AUDITOR — Full report: all advancements since RTA v17 — 2026-11-09 18:16

Role: Auditor, the audit-only keeper. Written Mon 2026-11-09 ~18:30 CST (America/Chicago). Created exclusively with noclobber, with a .sha256 sidecar and a PDF rendering. Governing direction: OW **c7795d6e88d5da7662d3a562f2564313ad472e0cb5560ce684cb067c61acd2b7** (OWNER_WRITE_VOICE_11fd320e_AUDITOR_FULL_REPORT_ALL_ADVANCEMENTS_SINCE_V17_2026-11-09_1816.md, sidecar MATCH, 18:16 CST), plus the voice amendment relayed by Physicist that the report also be rendered as one PDF. Scope:

- Local Path: Bridge C, A1→Ξ→RTA.
- Non Local Path: EH 11D, reverse-parity PT, non-local QM.
- The Auditor's full-corpus audit. All three point toward RTA and the geometric bridge in S^2-11DM^2ET-X. Read-only throughout. No existing file was edited, moved or deleted, and the v16 and v17 records are untouched. No CERT, no Gemini, nothing sent outside. Labels used: **PROVED / CONDITIONAL / OPEN / REFUTED**, plus PENDING where a record says so. Owner **DEFINITIONS** and decisions are not proofs. Pasted or Gemini-style text is not an Owner decision unless the Owner adopted it in their own words. Every file cited below was re-hashed this turn. An 8-hex prefix in the text stands for the full sha256 given in the ledger (§3) or in the earlier reports.

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0. Executive summary (plain language)

- **Baseline.** The RTA v17 DRAFT (2026-11-05 20:37 CST) has 156 pages and 74 printed Theorems; both counts were verified this turn with poppler (§1). Its old sidecar still disagrees with the PDF (custody item C17-1). A new dated sidecar written at 15:14 matches the PDF, but the cause of the old value is still unexplained.
- **Vacuum: an 11D background is now adopted, as a working choice.**
  1. The first proposal (D), $AdS_5 \times S^4 \times \Sigma_g$ (7708ec5d), was shown BF-unstable and dropped (822cecb2).
  2. Three supersymmetric twisted vacua were compared: MN N=2, MN N=1 and BBBW ($z = \pm 2, \pm 3$).
  3. After two holds (aaadb280, 587eca0c), MR2's full-11D file and PP's gate audit (384096a8) found no mathematical blocker.
  4. At 18:06 the Owner lifted the hold and **adopted MN N=2 as CONDITIONAL** (19a88b19). It is a working vacuum, not a proved one.
  5. Γ = **Bolza** (genus 2) was adopted for Σ (7ff8299d). Its systole, 2 arccosh(1+√2), is PROVED (computer-assisted).
- **Geometry definitions now on record (Owner DEFINITIONS, not proofs):**
  - brane = a stack of wrapped M5-branes, and face = the 10D matching hypersurface (f283cffd);
  - the face is a $\mathbb{Z}_2$ fixed plane, a Hořava–Witten-type wall (f825dfff);
  - Ξ names the curve $49a1 \otimes \chi_{140}$ (f283cffd, refined by 7b5a725b);
  - Λ_0 and R_c := $\kappa\ell$_P (5a6e7177);
  - P_rev = T1 and G = ½T^(EH) (2d1e877e).
- **O1 (the $\mathbb{Z}_2$ parity of G_4) is resolved as ODD by an audited derivation**, given the fixed-plane premise and premise P-b (PP 544fc7b0, 1bd92da7; OW f825dfff). Even parity on a fixed plane is REFUTED. The pasted parity-flip argument (record fe8bcb00) was **not adopted**, and its claim is REFUTED.
- **The junction is still UNSOLVED.**
  - Pure-M5 and E_8-wall sources cannot close it. The fibre-NEC no-go is PROVED at the end faces and CONDITIONAL in the interior.
  - A count gap N_req/(2N_flux) ∈ (1.255, 1.667) remains. It is CONDITIONAL, and BPS instantons leave it unchanged.
  - The HW wall's tr R^2 term (Y16) has the right sign, but only at O(1/N), so the leading-order verdict does not change. NL derived this; PP has not yet audited it.
  - Non-BPS E_8 existence (Y14) and the fibre equations with traceless stress (Y15) are OPEN and in progress.
- **MR2's wall curvature (bd47ff5e).** The bulk next to the wall is bounded. In the thin-wall limit the quadratic invariants diverge as $1/\varepsilon^2$, but they stay inside the EFT for any wall at least Planck-thick (CONDITIONAL).
- **Citation scope was widened twice** (3f160803, then 36683379). Any source may now be cited, provided its math is verified. A citation alone never makes a result PROVED.
- **Still neutral by Owner order:** the joint carrier, R_c's physical role, DP-a, Layer2 and any baseline (5f59e33e, amended only by f283cffd).
- **Still OPEN:** P5a (literal text does not type-check; its reading and referent are unanswered) and b.
- **Bridge C is PARTIAL.** Local has new evidence logs, but no Local FINDINGS file has been filed since 15:20.

- **RTA v18.** A **safe outline** exists (e18d8fea, 16:52). It holds algebra and definitions only and is not a draft. Earlier blockers 1 (vacuum) is now resolved as CONDITIONAL. Blocker 2 (junction) still stands, so the geometric v18 is still blocked (§7).

1. Version-17 baseline (stated first)

item	value	how verified
Path	reports/papers/RTA_FOUNDATION_BOUND_v17_DRAFT_2026-11-05_2037CST/RTA_FOUNDATION_BOUND_v17_DRAFT.pdf	on disk
Written	2026-11-05 ~20:37 CST (NOTES_v17.md); PDF CreationDate Thu Nov 5 20:46:59 2026 CST	NOTES_v17.md; pdfinfo
sha256	9b407e1b0805b2129ce95665b8c2343f9b7ccaa564126277c41b545bfff9d285	re-hashed
Pages	**156**	**VERIFIED**, pdfinfo
Printed Theorems	**74**	**VERIFIED**: pdftotext, distinct "Theorem N.M" labels at line start = 74. This matches audit_theorem_count_2026-11-05/AUDIT_SUMMARY.md and v17_all_printed_theorems_74.csv (74 rows).
Old sidecar .pdf.sha256	ee88d1cd57892bafff707e53517df30b81192a5988938ea6159d2dd78733763a	**MISMATCH** (C17-1), unchanged
New sidecar .pdf.CURRENT_2026-11-09_1514.sha256 (file sha 4ccbbb56…)	content 9b407e1b…	**MATCH**. Per OW ec97f45e (t4294): "empirical update; the cause of the old ee88d1cd is UNEXPLAINED. Old sidecar and PDF unchanged." Note file SIDECAR_NOTE_C17-1_2026-11-09_1514.md has sha 5dd223cf….
v17 content	v16 spine + post-v16 ledger + boundary rank G = $C_2 \times C_4 \times C_6$ + withdrawn appendix E	NOTES_v17.md
v16 (frozen)	papers/RTA_FOUNDATION_BOUND/RTA_FOUNDATION_BOUND_v16.pdf, 1fe7b8a2e245e90009c41726db13518b4c752e99ba4e08b4ffd006d8a73c3297	**MATCH** with its sidecar

2. Earlier Auditor reports this builds on (re-hashed, MATCH)

- AUDITOR_ADVANCEMENTS_SINCE_RTA_V17_REPORT_2026-11-09.md: 3c2c8fdb34f8b8cef4f133a331f60e5842c72ce65a7d9f702db752014018f811. Covers 12:30–15:05.
- AUDITOR_RTA_V18_READINESS_RECOMMENDATION_2026-11-09.md: 59da91863535b94bdcdd74b6bf103745efab15d270e4615d6e518b3bd4b4aea8 (sidecar MATCH). Covers 15:05–16:48.
- Full corpus audit (Auditor's lane): batches 01–23 (12:38–13:56 CST; 36 records with zero hash drift), errata ff9e4da9 ($\kappa/\Lambda_0/3\hat{}539$; 208 files, 606 line hits), crosscheck 034605bf (Artin v1 line 14b REFUTED; $W_{49}(f \otimes \chi_{-7}) = -1$) and final summary 0504d5d7 (corpus of 36,250 files / 22,035 unique; no tampering).

3. Chronological ledger (file time CST; full sha256; SC = sidecar check)

OW = Owner write, recorded by Physicist. Owner decisions came through the Physicist relay and were not confirmed in the Auditor chat. "rec only" marks a record that was **not adopted**.

3a. 12:30–15:05 (details in 3c2c8fdb)

time	record	sha256	SC	what / status
12:30–12:54	PP / Local / NL Soft SF audits (Ξ–Fricke–$\Gamma_0(49)$, SO_{48} v=23, P_rev/T^(EH) defs), Gemini question set and Physicist's Gemini check	see 3c2c8fdb §3	MATCH (Gemini question set: no SC)	Gemini answers are **not Owner decisions**
13:00	OW write/derive authorization	33d25a790fb1e0da5d93dc5cfb911963fa9a9a83c6dab6dfc8326c34096d7e15	MATCH	grant
13:20	OW B2 (B)-M + PP derive	68a2b8e1f14e872204da8aed62f895f5bf3315ef1ccdc04d4f14c3d81c04c09e	MATCH	B2 **CONDITIONAL** on (B)-M
13:23	OW B11/B12/B13/B10	45aeba5fc0433c808957d1518161cb24664ac97adff112d9df2ff7249a331ab3	MATCH	$\kappa := \sqrt{(2/\Lambda_0)}$, $\Lambda_0 = 0.783577$ **DEFINITION**; P5a OPEN
13:28–13:57	PP derive R_c / 11D map / metric / Israel / $AdS_5 \times S^4 \times \Sigma$_g params	9ea1a7ad…, 9dde377d…, 4d07bd1b…, 52acc3d7…, 4660b73e…	MATCH	EH flux lock REFUTED; R_c OPEN
13:40	OW C15-3	63deb8505bb1f53b4ad04c87b5a1824afda8b7526eb3322ab9f65511432f09b1	MATCH	κ supersedes $\Lambda_0 \approx 0.7858$
13:41	OW B16-1	0deba43db59d11d8c725eb5c643129f4be33be42d06931c3dae594abd6fc81ca	MATCH	**SUPERSEDED** by c4c0f1c0
13:42	OW B18-1	c4c0f1c0318691c045b136c01c4de7eb86690ae9c8db4d20b07939a6fd8b24ed	MATCH	$W_7 = +1$ on 7.4.a.a; **Path C stands**
13:51	OW F4 / errata / (D)	7708ec5d1518e371540f7b37a51db1c014e36563c458181cf692b50124a0edd7	MATCH	(D) adopted CONDITIONAL; replaced at 16:09
13:58	Ops F4 FINDINGS	d73cc7b1b9661cc3ced4255e02aba85fad64774e29091947aabab7b644913ee7	MATCH	parity + 3-bit status, exhaustive at 8 trits
13:58 / 14:04	Physicist Artin v1 / v2	65b367ef88cce23047c70ec0146bfb457c1274af2a3906f573d9a73d19b4b5b9 / ccb568287c9f08c5f976249a5ddf14b710efbeb36b7daf48ef2f9c0a01b826c0	MATCH	v2 fixes line 14b; not adopted
13:59–14:03	Auditor errata / crosscheck / final summary	ff9e4da9… / 034605bf… / 0504d5d7…	Auditor files (no SC by rule B10-1)	see §2
14:02	OW Ops HW sprint	adf7d966d233a74b900272bf3ff3227198945787261b66a738d4f5259b636077	MATCH	Gemini's PCIe / dual-rail / "ALL STOPS LIFTED" **not adopted**
14:05	PP derive twist $7.4.a.a \otimes \chi_{-7}$	90c1e058c6c7d2051b860c8f48b60ffe44265a6d2dd75288bf22108c2a67e63f	MATCH	$W_{49} = -1$, so not a P5a referent (PROVED)
14:08	OW P5a level-49 referent	e494873929ae6a499578a52e11e5c5bb18f1b40d14f7af2d9a42b9c78fe6004b	MATCH	referent g_0 (2,7,7), Fricke +1; P5a OPEN

time	record	sha256	SC	what / status
14:34	Ops HW sprint FINDINGS	2c3e5c18b6b48377b0f4a89a1ad77a3655411ce40d4c4215def2fceb30fe1172	MATCH	16-cycle design; formal only for N ≤ 32
14:54	OW R1 overflow rule	6b8822d2c09748f6545efbd4223de0a3e73f718c9a49907bd078103c5dfb28b7	MATCH	R1 adopted; sprint files only
15:01	OW P5a ρ̄	ea2cc9e6b41cde91844b4f2217d98694a8307df96a3a89bdd35d6eb291e46911	MATCH	ρ̄ = reducible mod-7 reduction of ρ_f; P5a OPEN

3b. 15:05–16:48 (details in 59da9186)

time	record	sha256	SC	what / status
15:05	PP record P5a referent adoption	05e1c40c4f14e6aace7d081b5649579615388f125cdb22827e031e6f92822872	MATCH	P5a OPEN
15:07	**PP derive P5a ρ̄**	7929ffe6535a79dff41ef6a8f6ca106566d29256b12fc56336ca63d8507d4b1d	MATCH	11 PROVED / 6 COND / 3 OPEN / 0 REF; e1: literal P5a text does **not type-check** (OPEN); e2/e3: vanishing REFUTED under PP readings I1/I2 (counted CONDITIONAL)
15:11	**OW blueprint (corrected)**	5a6e717732c413a39c89d9b95c9b17455ecf4da8dd64e8bb9d879f427e6a845c	MATCH	**DEFINITIONS** $\Lambda_0 := \ln(6+\sqrt{35})/\sqrt{10}$, R_c := κℓ_P; DP-Rc not adopted; Local/NL go-aheads; N = 4880, g = 2 as inputs. Blueprint text itself is Gemini-style, so only the widget answer counts.
15:13	OW t4294	ec97f45eeb84d62e7c4b0fa16c9cb133297d42a442a376f60707eb38fc4c22cc	MATCH	C17-1 dated sidecar; KK go; 324-trit design rests on simulation only
15:14 – 15:17	MR1 S2–S4 close + PP audit + amend	48a2d687..., 66a69888..., 0ea030ae...	MATCH	Appell→R_c route still OPEN
15:15	PP R_c = κℓ_P vs L_AdS/2	1ab88ebd04d671d7e8b5ee1015d3fbd5694e0c53814dae6f86b6a3e09096f3a5	MATCH	mismatch "forces nothing" (PROVED)
15:16	Ops R1 successor FINDINGS	060fd77e67c9a718a377b45b3ce2c23bbd0353a019e78a950a51a8698e59450c	MATCH	exhaustive N = 2..9 and 8 trits; formal 2–32 trits; **no 324-trit proof**
15:18 – 15:20	Local O3 SO_{48} datum, O5 intertwiner ι, O1/O2 Weil carrier, O7 Cand-1^+	d2d2d226..., ac1dd4e2..., ae951f3e..., 24e31966...	MATCH	PP audits below
15:20 – 15:32	NL F1 (T^(EH)), F2 (P_rev on −U), F3 (11D→Φ/W_{35}), O4 KK; MR2 map proposal, KK of (D), crosschecks	f2e5120d..., e9bdd871..., 965de066..., ceb85ec3..., fd191d28..., 7824e789..., 9354fe2a..., 3d943367...	MATCH	(D) BF-unstable
15:25 –	PP audits: Ξ Weil carrier 389c2f7c..., O3 3b6f8072..., O5 2c4269f2..., O7 6dc9a17d..., MR2 map	as listed	MATC	T^(EH) = 2T_std PROVED; $P_{-}^2 = -1$ PROVED; Cand-1^+ OPEN

time	record	sha256	SC	what / status
15:31	2b0a4125…, NL F1 8be401ed…, NL F2 c9c6ed65…		H	
15:29	Local relabel (ce4d9e80)	ce4d9e80…	MA TC H	contains a symbol swap; fix ordered at 17:30 (ff8a1670)
15:53	**OW P_rev = T1, G = ½T^(EH)**	2d1e877e292832e905e1b566f6d89eb589e96c27512fc8e0756bb5a254edb58a	MA TC H	adopted conventions; pasted t4311u/t4314u not Owner text
15:54	OW t4315 addendum	814625971f0bf84789b18da60b5ae4d07ccbbc4c859045db91d7d27b68aaeffa	MA TC H	4 pasted-source errors **not adopted**
16:03	PP audit MR2 KK/BF	9b557373954788a89d5c44b8eac6d09f1c5c131efdb7386464d85022a9c869fc	MA TC H	(D) unstable (14 at −16/3 < −4)
16:09	**OW replace (D)**	822cecb27e974268175d5d88d6b0ab032fc7be1d85ea5f10321d9c24da128ca0	MA TC H	supersymmetric twisted vacuum; compare MN2/MN1/BBBW first
16:19	NL twisted-vacua compare	418e4e50e4ce791bfb7c00fe4916dcee7082027c38ed7c55fea82659c98838e6	MA TC H	—
16:30	OW t4327 (adopt pasted text)	ed526f70bf045f86f21fb39977dbebf722dedfb3f859ec1e56a24b07f881b882	MA TC H	**WITHDRAWN** at 16:31
16:31	**OW neutral state**	5f59e33e5b49ed7f7a89b06f0ceff0b7bc7dc034c83d5c65a72bea176aa3a234	MA TC H	everything unresolved and neutral; Λ_0/R_c definitions kept
16:36	PP twisted-vacua audit	df503cd99df53b222b27f5d6a2dfbb1fb09dd9c7237630d72a7fff36a75ed494	MA TC H	all viable; recommends MN N=2; MR2 §6 curvature check broken
16:40	**OW vacuum HOLD**	aaadb280c54dac3f9a9b131435092511396b95a43b15333c389965f26c395585	MA TC H	until MR2 fix + joint table + PP
16:44	**OW brane / face / Ξ**	f283cffd365ca7368a9f7b5dfdca7ee177ed54db25a28c1944999a5af755e5b6	MA TC H	**DEFINITIONS**; rest neutral
16:45	NL / Local carry-forwards	6549982d… / 60778d10…	MA TC H	—
16:46	**OW NL junction GO**	4810280562b2fb0cf94de47b9a92c840643a7dac077e500d26bfa644880f7d24	MA TC H	three candidates; b OPEN

3c. 16:48–18:17 (new in this report)

time	record	sha256	SC	what / status
16:51	Auditor v18 readiness	59da91863535b94bdcdd74b6bf103745efab15d270e4615d6e518b3bd4b4aea8	MA TC H	hold the geometric v18; outline optional
16:55	NL junction setup on constant-radius faces	3cd6c8d303b3e07a8bd7d5de3adb9ff6b53f95d217b7d4a160431bdecf07c050	MA TC H	P3a: smeared-M5 Israel on the exact end faces **REFUTED**
16:56	**RTA v18 SAFE OUTLINE** (algebra + definitions only)	e18d8fead184dd0e5fc04fa21422fdb3b7c6fd6c5544160f2a9bbd7cbb3b633a	MA TC H	outline, not a draft, nothing frozen; excludes vacuum, junction and geometric Bridge C
16:59	PP audit of NL junction + Ξ Fricke	5cbd050abc2fe085f715d548fffb58b202d7c685a54070c288972b5150cb0fcb	MA TC H	12/6/4/0 PENDING/1 REF. No-go only in a narrow scope. Ξ: conductor 19600, root number −1, w = +1 (PROVED); Ξ's newform = the −20 twist's newform
17:01	NL erratum 5	59218e611509a3ab4d6e5cc593c51d39d84fdb52cb84e8dd50914372ff8c8d11	MA TC H	accepts 5cbd050a
17:03	OW voice t4352 record	1cb3b8d524f4691860f9c1f8f7cb8d263d1bdc55137519fe1378ac43cff143a3	MA TC H	its "even O1 by definition" item was **WITHDRAWN** at 17:03
17:03	MR2 twisted-vacua compare .md	98e502c37d42c7dc1dde4e79ce383faac14ce631545f25c50b3b601d0afa524d	MA TC H	—
17:04	**OW Ξ curve, NL templates GO, O1 OPEN**	7b5a725befe67c64a52878a348190a20cda891b8c0618e51a47820cd7c76010d	MA TC H	Ξ := the **curve** 49a1⊗χ_{140} (isogenous to the −20 twist); O1 OPEN
17:04	PP audit NL erratum 5	cb4d26b06d4205dccf7f714a8d38523434d02d43d464caf912a9067f33db9740	MA TC H	P3b PROVED given J1
17:04 – 17:06	JOINT MR2+NL table; tally erratum; face addendum	0b364d6e626f71f80e763eb8ccd504f512ae65f5187c8428d12464619c59f3e4; dc9dd1a0…; 97d671e3…	MA TC H	joint table landed
17:05	NL erratum 6 (citations)	5c012d06…	MA TC H	—
17:05	**OW O1 even only by derivation**	9a8ced02a7053846d9f91c6600ad01e3c08673162045d5d910f5e4889c5757ba	MA TC H	no bare definition; O1 OPEN
17:11	NL face stress-energy templates	53096d433d062f51e0fc078d4bdd5faee333cbcf7edadd66c2a0b3fd4d2e7fb7	MA TC H	—

time	record	sha256	SC	what / status
17:13	NL non-constant faces / localized M5 / O1 attempt	e2065e05c7d552117895351482cbb3838fcd651bb8873cd6398d0e35b12fd389	MATCH	—
17:30	**PP audit f3c5d73f** (templates + non-constant faces)	f3c5d73f371b31a75a87021452251af77b9d2d37e7fb6b955f647ed84192d523	MATCH	31/21/5/4. A pure-M5 face needs fibre-NEC-violating stress (PROVED at end faces, interior CONDITIONAL). NC1 over-claimed (twisted-φ Lie-drag REFUTED). Count ratio ∈ [1.255, 1.667] CONDITIONAL. PD1 PROVED.
17:30	OW pursue all open items, all bots	ff8a16701cd0832fc449c91fc9b2a2dc4ff84bf743a4fd757e07d21ddbacd441	MATCH	per-bot assignments
17:35	PP EFT re-audit (MR2 V2 + joint table)	fc5e0156512bcbdd3b538cd6bceceaaafed959587c85589d0d41b8e44509aeb6	MATCH	all four candidates CONDITIONAL, none REFUTED; $\ell_p^2 R_max$ 2.8e-3–6.7e-3; absolute systole OPEN; MN2 recommendation stands
17:35	NL erratum 7	130e27969e0d5921029007475c185c9006d5e510abd7ec6b614e6a58d4974c71	MATCH	accepts f3c5d73f
17:36	NL O1 resolution pass	13ccd16251d055aa735dac4be75bc8e9f90d95d087ea0cff8b118efb80965f59	MATCH	on a $\mathbb{Z}_2$ fixed plane O1 is forced ODD (given P-a/P-b); even REFUTED within P3
17:36	**OW record t4363 (pasted parity flip)**	fe8bcb00e623f7ad0e1174b48db7e78fdd52b66cdb0ea63c7ad663da394aa131	MATCH	**rec only, NOT adopted**; 7 conflicts flagged
17:37	**OW vacuum HOLD (extended)**	587eca0c108f39b15c8d2f372861087ed03796dc28f9a49449f95db0dead7fcf	MATCH	until MR2's full-11D file + PP audit
17:38	OW conditional flip plan	602b6427feeab3225ebac343be5d60dfb69de61a78fa2090ba1fc7e3235acb13	MATCH	**not a decision**; superseded by f825dfff
17:43	**PP audit 544fc7b0** (pasted flip vs NL O1)	544fc7b0b0183022ad1d201933d28de3db9dc58d2a5a7e0b9103bf8675b6450c	MATCH	16/7/5/7/1 PENDING. A fixed plane forces ODD. The pasted anomaly argument is not sound; "REQUIRED" REFUTED; the "PP 17:02 audit" attribution REFUTED
17:45	NL erratum 8	96fb60698c4f0dc59c1f27b98beb6a2cf3df47d08141956760ea2707636d80cd	MATCH	accepts 544fc7b0
17:46	**OW face = $\mathbb{Z}_2$ fixed plane (HW wall), O1 odd**	f825dfffb8cd17a7f3f57a2b0a70bd0d9c9d73f215febffdbd2ca95b494d7cd1	MATCH	Owner **DEFINITION** of the face; O1 ODD via the audited derivation; E_8 sector CONDITIONAL; N_M5 + n_inst = 2N_flux
17:48	PP audit erratum 8 + OW f825dfff	1bd92da71029f93c2800cb2dc5c131a7b52010315308f0e7f9c8b175090427bc	MATCH	14/3/1/1/1; 5 of 6 relay points match, the sixth is not in the record
17:48	**MR2 full-11D EOM / Riemann / systole**	9ab8f0a6c25ba5e46b3eb15a3734e6a0e98573eb07b992e12b474c04a8905bb6	MATC	gate file

time	record	sha256	SC	what / status
			H	
17:48	MR2 erratum (per PP EFT re-audit)	2efef556f4080740380701eb77ec2bceeacd2710804eca8b70cea5ade9dcac52	MATCH	6 fixes applied (PROVED by 384096a8)
17:50	**NL odd HW wall, E_8 sector**	4d9c9951eedc43b359d63842ad1535084b43bf893eca3125a4c7b06c0014468a	MATCH	E_8 cannot close the junction; BPS count gap unchanged
17:50	NL erratum 9	76ec4e75d73c12700c038b8eb00d047df85148f789c3f9d2d375ad50890aba1f	MATCH	O1 wording
17:53	**PP audit 7f0ce656** (erratum 7 + NL 1736 no-gos) + rollup self-erratum 724325dd…	7f0ce656ebecc94258bd06cb412560b334d10cc5e825b25a523460214b363c4d	MATCH	32/6/6/2. B1/B2 PROVED given J1 for every z. "No face point totally geodesic" REFUTED for general faces
17:54	**OW queue jobs + Γ = Bolza**	7ff8299dbcaf3b526936b40a9e24d8b9e58edf13fd9a9a93e822a92ca82a16ac	MATCH	NL E_8 stress job; MR2 wall curvature job; Γ := Bolza. **Pasted t4377u NOT adopted** (7 items)
17:57	**PP audit bf4773a8** (NL 1750 + erratum 9)	bf4773a80192a5d34905237c5a3942aa2c601af6f3189c479d9d6f80e1b63d73	MATCH	22/16/4/0. "All sources obey the NEC" CONDITIONAL on dropping the tr R^2 term
18:00	**NL E_8 instanton stress on the odd HW wall, Γ = Bolza**	859328a2003242ca59109ffb699ba69995f4d5c9dec0055e49d16adc4e2d379d	MATCH	Y4: no E_8 field supplies the fibre-NEC deficit; Y10 gap unchanged; Y14/Y15 OPEN
18:01	NL errata 10, 11	52424877… / c2864c22…	MATCH	accept 7f0ce656 / bf4773a8
18:05	**PP gate audit 384096a8**	384096a8581c1b29bb59768877f36f7fd6caa10f72a23528ea8d058e1253cc19	MATCH	16/13/3/0; release condition of 587eca0c met; no mathematical blocker
18:05	**OW pursue NL E_8 open paths**	9bb1858ce34ee985d18d5a1db3ca838f20e2fddab741d7732e970e1e354ff27c	MATCH	Y14, Y15, Y16 (no ordering instruction)
18:06	**OW lift HOLD, adopt MN N=2 CONDITIONAL**	19a88b1958753ecaaa999e0ce763bca7fd449069278dd4bdb87cea002613ee03	MATCH	adopted working vacuum
18:07	**PP audit fe0c3d92** (NL 1800 + errata 10/11)	fe0c3d92fdfac839653f9358f0a2c193c91a9e9dd42d46bbcbc6d58e1b794ae3	MATCH	44/11/3/3; equality form only for magnetic F; "Y16 first" REFUTED by quotation
18:11	**MR2 wall-boundary curvature**	bd47ff5e8f62a6226680a93107cacae8c8be573399e1f101a2dc57635f03fbfb	MATCH	bulk bounded; thin wall diverges as $1/\varepsilon^2$

time	record	sha256	SC	what / status
18:11	MR2 erratum to bd47ff5e	2368ffbdc6cd6bbe9482e476a085f5d3193da21ef1ea5e2324d3a4a44bf6f295	MATCH	"warping lengthens systole" REFUTED → **OPEN**
18:13	**NL Y16** (tr R^2 wall stress)	7270a6b35c97822a4efe37b7f6e78d94efafd0ae44d298ba70554e926c9d5237	MATCH	NEC-violating sign, but O(1/N); verdict unchanged at leading order. **PP audit NOT FOUND**
18:13	NL erratum 12	53d4f945e2d4df565cfa7cdae90dcda8a7b8c0c33855ee3efdf79004cd682230	MATCH	accepts fe0c3d92
18:14	**OW citation scope widened**	3f1608039bc68b62332a07a7621da749ca303f690993048bef5133282fbc9320	MATCH	beyond arXiv, incl. AI-assisted research
18:15	**OW citation scope: all sources, math checked**	36683379e18ac23c46236b37ed45ec1ddf70a93ac9094d2abc476c953433904f	MATCH	amends 3f160803
18:16	**OW this report**	c7795d6e88d5da7662d3a562f2564313ad472e0cb5560ce684cb067c61acd2b7	MATCH	governs scope
17:36 – 18:17	Local evidence_local_bridgeC_resolution_2026-11-09/ (part1, part3, part4 .py/.log)	no sidecars	—	evidence only; **no Local FINDINGS .md**, not audited
17:36 – 18:17	MR2 symbolic-Einstein / global-curvature / Maxwell / per-point scripts and logs	some without sidecars	—	in progress; no .md for the symbolic/global job yet

4. Advancements by topic, with status

4.1 Vacuum: the four vacua and adoption (NL / MR2 / PP)

- **(D) $AdS_5 \times S^4 \times \Sigma_g$.** Adopted CONDITIONAL at 13:51 (7708ec5d). It is **REFUTED as a stable vacuum**: the SO(5) 14 has $m^2 L_5^2 = -16/3 < -4$, and the probe-M5 ratio is $\sqrt{6}/2 > 1$ (7824e789, 9b557373). Replaced by OW 822cecb2.
- **The comparison set** (df503cd9, joint table 0b364d6e, fc5e0156, 9ab8f0a6, 384096a8):
  - MN N=2 (z = ±1, 8 supercharges);
  - MN N=1 (z = 0);
  - BBBW z = ±2;
  - BBBW z = ±3. With (D) as the reference, these are the "four vacua". All four are CONDITIONAL; none is REFUTED.
  - Existence and the 11D uplift are CONDITIONAL on citations (BBBW 1203.0303, MN hep-th/0007018, CLP).

- BF stability is CONDITIONAL (SUSY unitarity).
- The M5 wall ratio is exactly 1, the BPS value: the identity $(X_1X_2)^2(X_1+X_2) = 3/2$ is PROVED.
- Central charges at N = 4880 are PROVED as arithmetic.
- **MR2 checks.**
- The §6 curvature check failed at 16:36: it was non-discriminating against its negative control. It was replaced by V2 and the full-11D file 9ab8f0a6 (erratum 2efef556).
- In MN2, Maxwell, G∧G and λ are PROVED identically. Einstein and Bianchi are PROVED at 29 sampled points including the poles; the all-points statement is CONDITIONAL.
- Curvature: $\ell_p^2|K|_{max} = 5.40e\text{-}3$ and $R^4 \sim 1.6e\text{-}7$.
- **Gate (PP 384096a8, 18:05).** The release condition of 587eca0c is met and no mathematical blocker was found. MR2 still owes 8 fixes (§14 of that file).
- **Adoption (OW 19a88b19, 18:06).** The hold is lifted. **MN N=2 is adopted as CONDITIONAL**, a working vacuum and not a proved one. Carried caveats:
- the all-points Einstein/Bianchi check;
- the KK spectrum (Wilson lines, spin structure) is OPEN;
- MR2's "warping lengthens the systole" claim is OPEN (2368ffbd);
- the BBBW dual is moot for the adopted vacuum but still OPEN.

4.2 Γ = Bolza for Σ

- **OW 7ff8299d (17:53): Γ := Bolza (genus 2).** This is an Owner adoption.
- Bolza's systole $2\,\mathrm{arccosh}(1+\sqrt{2}) = 3.05714183896$ is PROVED (computer-assisted, via the Poincaré polygon theorem; 384096a8 §10).
- The systole for MN2 is $\geq 12.1 \times 2\pi\ell_p$ (PROVED).
- That the systole is the global genus-2 maximum (Jenni 1984) is cited and unverified, so CONDITIONAL (NL erratum 12). It is to be re-checked under 36683379.
- Not fixed by Γ: the flat E_8 / Wilson-line moduli (real dimension 496, PROVED, Γ3) and the spin structure. These affect the charged KK spectrum (OPEN).

4.3 Face, O1 parity and the HW wall

- **Definitions (Owner, not proofs):**
  - brane = a stack of wrapped M5-branes, and face = the 10D matching hypersurface (f283cffd);
  - the face is a $\mathbb{Z}_2$ **fixed plane** of HW type (f825dfff).
- **O1, the parity of G_4:**
  - 16:44–17:05: O1 OPEN. 9a8ced02 says even parity may be locked only by an audited derivation. The "even" item of voice record 1cb3b8d5 was WITHDRAWN (7b5a725b).
  - 17:36: NL 13ccd162 derives that a fixed plane forces ODD, given P-a and P-b.
  - 17:43: PP 544fc7b0 confirms it:
    - P-fix forces ODD, and both even and "flip" are impossible on a fixed plane (REFUTED);
    - "flip ⇒ even" holds only in the non-fixed case P-nofix (i), where it costs zero net face charge.
  - 17:46: OW f825dfff records **O1 = ODD**. It is PROVED given P-b, under the fixed-plane premise; P-b itself is a CONDITIONAL premise. PP 1bd92da7 verifies the record.
- **Consequences:**
  - face charge N_M5 + n_inst = 2N_flux = 9760 at N_flux = 4880 (CONDITIONAL, HW);
  - an E_8 wall sector is required (CONDITIONAL, HW).
- **Not adopted:** the pasted parity flip (fe8bcb00, t4363u), whose claim PP REFUTED, and the conditional flip plan 602b6427, which was superseded.

4.4 Junction, E_8 wall and the count gap (NL / PP)

- **Setup** (NL 3cd6c8d3, 16:55; PP 5cbd050a). On constant-radius faces with a smeared-M5 source only, Israel fails on the exact end faces (P3a, REFUTED). This is not a general no-go.
- **Templates and non-constant faces** (NL 53096d43, e2065e05; PP f3c5d73f, 17:30):
  - Constant-face no-go: a pure-M5 $\mathbb{Z}_2$ face needs fibre-NEC-violating stress, and no positive-tension brane mix can supply it. PROVED at the end faces; CONDITIONAL in the interior (dense numerics, reproduced by PP).
  - "w_fib ≤ −1.2" is REFUTED (counterexample −1.1925), and the B1 printed sign is REFUTED.
- **General faces** (NL 13ccd162 §B–D; PP 7f0ce656, 17:53):
  - Static C^2 faces that attain a minimum of r have no solution: B1/B2 PROVED given J1, for every z.
  - "No face point is totally geodesic" is REFUTED for general faces and PROVED for constant-r faces.
- **E_8 wall** (NL 4d9c9951, 17:50; PP bf4773a8; NL 859328a2, 18:00; PP fe0c3d92):
  - The Yang–Mills stress obeys the NEC (E2/Y3, PROVED). So **no E_8 field, BPS or not, can supply the fibre-NEC deficit** (Y4: PROVED given J1 at the end faces, CONDITIONAL in the interior).
  - The identity $K(\hat{e},\hat{e}) - k_t = -(\kappa^2/2\lambda^2)|F(\hat{e},\cdot)|^2$ holds only for magnetic F. "For any E_8 field" is REFUTED; the inequality used holds for every field.
  - BPS instantons trade one-for-one with M5s, so **the count gap N_req/(2N_flux) ∈ (1.255, 1.667) is unchanged** (Y10, CONDITIONAL). Earlier "5/3 gradient" and "(2489, 6510)" readings are not adopted / REFUTED (7ff8299d; erratum 12).
  - Non-BPS E_8 energy could at most close the count gap, never the NEC deficit (CONDITIONAL).
- **tr R^2 wall term** (NL Y16, 7270a6b3, 18:13):
  - The $-\frac{1}{2}\mathrm{tr}\, R^2$ stress has the NEC-violating sign at the UV end (Q4b, CONDITIONAL on c > 0).
  - Its integrated supply is −2 M5 units (Q5), an O(1/N) effect relative to the deficit.
  - So **the junction verdict does not change at leading order in 1/N** (Q11a, PROVED given J1).
  - The interior sign (Q8) is OPEN. **PP audit of Y16: NOT FOUND** (PENDING).
- **Y14, non-BPS E_8 existence** on the squashed, twisted, warp-weighted fibre: **OPEN, pending** (9bb1858c, ETA 3–5 h). Cited inputs: Sadun–Segert, Taubes, SSU, BLS. These had been marked unverified under the old arXiv-only rule and are to be re-checked under 36683379.
- **Y15, the fibre Israel equations with traceless stress: OPEN, pending**. It depends on Y14 (ETA ~2 h after Y14).
- **Net: junction UNSOLVED for MN N=2** (19a88b19). b is OPEN. J7 is OPEN: no extra source is adopted, and the pasted "J7 REJECTED" was not adopted.

4.5 Wall curvature (MR2 bd47ff5e, erratum 2368ffbd)

- **Bulk limits on each side: BOUNDED.** PROVED at sampled points; the global maximum is CONDITIONAL (dense scan). Riem^2 max = 3407.98.
- **Thin-wall limit of the smoothed profile:** Riem^2 and Weyl^2 diverge as $1/\varepsilon^2$ (PROVED in-repo). The wall is non-umbilic.
- **Physical size:** $\ell_p^2\sqrt{\mathrm{Riem}^2} \approx 0.101$–$0.114$ at $\varepsilon_y = \ell_p$, so the wall is inside the EFT for any wall at least Planck-thick (CONDITIONAL on that thickness premise).
- The fibre-NEC deficit is independently reproduced. The wall energy sits inside the count gap; the anisotropy is what fails.
- Erratum: "warping lengthens the systole" is OPEN, neither PROVED nor REFUTED. The bound $\mathrm{sys}_{11} \in [75.95, 85.25]\,\ell_p$ is PROVED.
- The MR2 symbolic-Einstein and global-curvature job (scripts 18:00–18:17) has **no .md yet** (NOT FOUND).

4.6 Citation-scope corrections

- **3f160803 (18:14):** widen beyond arXiv to include AI-assisted research not on arXiv. Flag prior findings narrowed by the arXiv-only rule.
- **36683379 (18:15), which amends it:** ALL sources are allowed, provided each cited result's math is verified. A citation never makes a result PROVED.
- Effect: items marked "unverified (outside arXiv allowance)" stay CONDITIONAL until their math is checked; no automatic upgrade. These are NL erratum 12 §8 and the PP fe0c3d92 compliance note (Jenni 1984; Sadun–Segert; Taubes; SSU; BLS; DOI/PMC/AMS/Wikipedia titles). **The re-check is not yet on disk** (PENDING).

4.7 Definitions and conventions in force (Owner; not proofs)

- $\Lambda_0 := \ln(6+\sqrt{35})/\sqrt{10} \approx 0.7835772$, and $R_c := \kappa \ell_P$ with $\kappa = \sqrt{(2/\Lambda_0)} \approx 1.597622$ (5a6e7177). Also on disk: $\Lambda_0 = \sqrt{(7/8)} \cdot L(1,\chi_{140})$, PROVED. The **dynamical derivation of Λ_0 and R_c's physical role are OPEN**.
- $P_{rev} = T1\ (P_- = (S \cdot \hat{P})|_{H_-})$, and $G = \frac{1}{2}T^{(EH)} = T_{std}$ (2d1e877e). $P_-^2 = -1$ and $T^{(EH)} = 2T_{std}$ are PROVED (c9c6ed65, 8be401ed).
- $\Xi :=$ the curve 49a1$\otimes\chi_{140}$: conductor 19600, Fricke +1, isogenous to the −20 twist, same newform and L-function (f283cffd, 7b5a725b; PP 5cbd050a PROVED).
- B2 is CONDITIONAL on (B)-M (68a2b8e1). κ is defined as above (45aeba5f, 63deb850). $W_7 = +1$ on 7.4.a.a, and Path C stands (c4c0f1c0).
- Inputs (not forced): $N = 4880$, $g = 2$, standard flux $(2\pi\ell_P)^3 N$ (5a6e7177). $\Gamma =$ Bolza (7ff8299d). Vacuum: MN N=2 CONDITIONAL (19a88b19).
- **Neutral by Owner order** (5f59e33e, amended only for brane, face and Ξ by f283cffd): the joint carrier is undefined; R_c's role is OPEN; DP-a and Layer2 have no stance and are not locked; there is no baseline.

4.8 Local Path: Bridge C A1→Ξ→RTA

- **Status: PARTIAL** (f283cffd; typecheck 6e7cd1c5…).
- **PROVED** (389c2f7c, 3b6f8072, 2c4269f2):
  - $W_{49} = -1$ on 49a1 and $S_2(\Gamma_0(49))^+ = 0$;
  - Ξ has level 19600 and Fricke +1;
  - $\Lambda_0 = \sqrt{(7/8)}L(1,\chi_{140})$.
- **CONDITIONAL:**
  - the induced-carrier step;
  - χ_{140} as the Weil character on all of $\Gamma_0(140)$. New numerics on disk are not yet filed as FINDINGS: part4 log, max residual 5.2e-13 over 48 Galois twists, with a control; this is supporting evidence only.
- **OPEN:** the joint-carrier level (980 vs 19600), with the joint carrier itself undefined.
- **Local evidence 17:36–18:15** (part1/3/4 logs; no sidecars, no FINDINGS .md, not audited):
  - Ξ's newform equals $f\otimes\chi_{-20}$;
  - $L(\Xi,1) = 0$ numerically;
  - the Kohnen plus space $S^+_{3/2}(196)$ has dimension 0;
  - $c(n) \neq 0 \Leftrightarrow L(f\otimes\chi,1) \neq 0$ for 75/75 squarefree $n \leq 120$.
- **Other Local results:**
  - O5 intertwiner ι: PROVED 6 / COND 1 / OPEN 1 / REF 1 (2c4269f2);
  - O7 Cand-1^+ ($n \geq 1$): OPEN (6dc9a17d);
  - O3 SO_{48} datum with the Codex π (3b6f8072);
  - the ce4d9e80 symbol fix was ordered at 17:30 (ff8a1670), but its **filed fix is NOT FOUND**.

4.9 Non Local Path: P_rev and non-local QM

- P_- is antiunitary, a particle–hole symmetry and not a chiral one. There is no Kramers degeneracy at $E \neq 0$ (PROVED, c9c6ed65).
- The R-symmetry versus P_rev interplay is OPEN (df503cd9 C3).
- There is no bulk→$H\otimes\mathbb{C}^2$ map, so no conflict at the level of definitions (CONDITIONAL).

4.10 P5a

- Referent g_0 with $(a_2,a_3,a_5) = (2,7,7)$ and Fricke +1 (OW e4948739). $\bar{\rho}$ = the reduction mod 7 of ρ_f, reducible, with $\bar{\rho}^{ss} = \omega\oplus\omega^2$ (OW ea2cc9e6).
- **PP 7929ffe6 (15:07):**
  - **e1: the literal P5a text does not type-check (OPEN).**
  - Under PP's readings I1 and I2, the vanishing claim is refuted by PROVED counterexamples: $F_{old}^+ = f(q)+49f(q^7)$, and the orbit-5 form h. These count as CONDITIONAL, since they depend on adopting a reading.

    - The non-semisimple refinement (b2) is OPEN.
- **P5a is OPEN.** The Owner's reading (I1/I2/other) and referent (g_0 / h / f(q)+49f(q^7)) are unanswered (ec97f45e). The earlier note that the all-n congruence rests on Chen–Kiming (CONDITIONAL) still applies.

4.11 Carried from the earlier reports (unchanged status)

- Artin conductor v1 line 14b is REFUTED, and v2 fixes it (not adopted).
- Errata ff9e4da9 (κ 1.595361, Λ_0 0.7858, 1.595436, 3^539 corrections).
- Ops:
    - F4 (d73cc7b1);
    - the 16-cycle sprint (2c3e5c18);
    - the R1 successor (060fd77e), which is exhaustive and formal only at small widths, with no 324-trit proof;
    - the negative floor-division extension F(n) is OPEN;
    - no Ops records after 15:16.

5. Status roll-up (headline items)

status	items
PROVED	Bolza systole; MN2 Maxwell/G∧G/λ identically, Einstein/Bianchi at 29 points; (D) BF-instability (arithmetic); M5-wall BPS identity (3/2); a-values at N = 4880; P-fix ⇒ odd (given P-b); PD1 sign rule; YM-stress NEC; Y16 Q5/Q11a (given J1); B1/B2 general static faces (given J1); wall-curvature $1/\varepsilon^2$ divergence; sys_{11} bound; Ξ Fricke +1, conductor 19600, newform = −20 twist; T^(EH) = 2T_std; $P_{-}{}^2 = -1$; $\Lambda_0 = \sqrt{(7/8)}L(1,\chi_{140})$; $W_{49}(f\otimes\chi_{-7}) = -1$; $\varepsilon_7 = +1$
CONDITIONAL	MN N=2 as the adopted vacuum and all four candidates (citations, all-points EOM, BF via SUSY); E_8 wall sector; N_M5 + n_inst = 2N_flux; count gap (1.255, 1.667); interior NEC no-go; EFT inside for walls at least Planck-thick; P-b premise; B2 on (B)-M; Bridge C induced carrier and χ_{140} on $\Gamma_0(140)$; P5a refutations under readings I1/I2; citations outside the old allowance (pending re-check)
OPEN	junction (unsolved), b, J7; Y14 non-BPS E_8 existence; Y15 traceless fibre equations; Y16 interior sign (Q8); KK spectrum (Wilson lines, spin structure); warping vs systole; BBBW dual; P5a (reading, referent, e1); Λ_0 dynamics; R_c role; joint carrier level; Appell→R_c route; Cand-1^+; α; Φ pointwise vs integrated; LD-O3a/b/c; H_+U; R-symmetry vs P_rev; Ops negative extension and 324-trit proof; C17-1 cause
REFUTED	(D) as stable; smeared-M5 Israel on exact end faces (P3a); "w_fib ≤ −1.2"; B1 printed sign; twisted-φ Lie-drag; "no face point totally geodesic" (general faces); even/flip parity on a fixed plane; pasted anomaly "REQUIRED" and the "PP 17:02 audit" attribution; Y3 "for any E_8 field" equality; Y15 interval (2489, 6510); "Y16 first" as Owner text; arXiv:2009.12051 attribution; and the earlier list (EH flux lock, T_3 isometric embedding, pure-tension face, EH flux normalisation, Artin v1 14b)
NEUTRAL (Owner order)	joint carrier, R_c physical role, DP-a, Layer2, baseline

6. Pasted / Gemini-style text NOT adopted

item	on disk?	status
fe8bcb00 (record of pasted t4363u: parity flip + bot directives)	YES, 17:36, sidecar MATCH	**recorded, NOT adopted**. Parity claim REFUTED (544fc7b0); "J7 REJECTED", "Warden confirmed" and "b solved dynamically" not adopted
t4363 (chat turn)	text only inside fe8bcb00 (no standalone file)	as above
t4371 (chat turn)	**NOT FOUND on disk** (no file or text mentions t4371)	**relayed only**; no status

item	on disk?	status
t4377 (chat turn; pasted t4377u)	no standalone file; rejected items listed in OW 7ff8299d §"NOT adopted"	**NOT adopted**: DP-a "rejection", "minimal joint carrier", "5:02 PM baseline", "5/3 gradient", "b solved dynamically", Warden lock, strategy-map claims. Only the two jobs and Bolza were adopted.
Earlier	blueprint text t4289u (5a6e7177), t4311u/t4314u (2d1e877e, 81462597), t4327 pasted text (ed526f70, withdrawn), t4333u (aaadb280), t4337u (f283cffd beyond its three items), Gemini 14:02 items (adf7d966)	not adopted beyond what each OW states

7. Blockers for a geometric RTA v18

1. **Junction unsolved** for the adopted MN N=2. The leading-order no-go stands. Y14 and Y15 are pending, and the PP audit of Y16 is NOT FOUND. b is OPEN.
2. **Vacuum adopted, but CONDITIONAL.** The all-points EOM, KK spectrum and spin/Wilson-line data are OPEN, and MR2 owes 8 gate fixes.
3. **Neutral items** need an Owner stance, or v18 must list them as OPEN: joint carrier, R_c role, DP-a, Layer2, baseline.
4. **The S²-11DM²ET-X objects are not finished:** the 11D map (Φ pointwise vs integrated), V_7eff and G_5 (on hold since 822cecb2, now unblocked by the adoption but not derived), and Λ_0 dynamics.
5. **P5a OPEN** (text does not type-check).
6. **Bridge C PARTIAL**, and no Local FINDINGS since 15:20.
7. **Citation re-check under 36683379 is pending.**

Status of the readiness recommendation (59da9186):

- Trigger (i)–(iii) are now met: MR2 fixed check, joint table + PP EFT re-audit, vacuum adopted (CONDITIONAL).
- Triggers (iv) and (v) are not met: no junction for the adopted vacuum, and no stance on the neutral items.
- So **the geometric v18 is still not ready to draft.** The safe outline (e18d8fea) may be extended with the new definitions (face fixed plane, Bolza, MN N=2 as CONDITIONAL) if the Owner wishes.

8. Carried-forward gaps

- Junction, R_c role, Layer2 transfer, S²-11DM²ET-X bridge objects.
- P5a; B7-1…4, B10-2, B10-4, B11-2, B13-1/2/4/5, B14-1…4, B19-4, B22-1 (see 0504d5d7 §7).
- C17-1 cause; B4-1, B4-2, B9-2, B9-3.
- Ops F(n) on negatives and the 324-trit proof.
- ce4d9e80 symbol fix not yet filed.
- MR2's §14 fixes (384096a8).
- NL fix lists in 5cbd050a, f3c5d73f, 7f0ce656, bf4773a8 and fe0c3d92 (errata 5–12 on disk accept them).

9. Awaiting the Owner

- Stances on the neutral items, or an instruction to list them as OPEN in v18.
- Whether to start the geometric v18 after Y14/Y15/Y16 and a junction result, or to extend the safe outline now.
- P5a reading and referent; α; Φ pointwise vs integrated; LD-O3a/b/c; H+U.
- Whether P-b (the $\mathbb{Z}_2$ symmetry, including the CS and I_8 terms) is accepted as the premise behind O1 = odd.
- Warden's three-hour cadence must be confirmed in Warden's own chat (fe8bcb00 conflict 4).
- Ops next steps (negative extension; 324-trit proof).
- Codex re-upload (carried).

10. Custody statement

- Every OW and finding cited in §3c, and every listed prefix, was re-hashed this turn. **All found, and all sidecars MATCH**, except where §3 says "no sidecars" (Local evidence logs, some MR2 scripts in progress, and Auditor files by rule).
- **MISMATCH:** only the v17 PDF against its original sidecar (C17-1). The new dated sidecar matches.
- **NOT FOUND:**
  - a separate MR2 "ERRATUM2" to gate file 9ab8f0a6 (the only later MR2 erratum is 2368ffbd, which applies to bd47ff5e);
  - any t4371 text;
  - a PP audit of NL Y16 (7270a6b3);
  - a Local FINDINGS after 15:20;
  - the ce4d9e80 fix;
  - the MR2 symbolic/global-curvature .md;
  - the citation re-check under 36683379;
  - any RTA v18 draft. Only the safe outline e18d8fea exists.
- v16 and v17 were not touched. Nothing was edited, moved or deleted. This file and its sidecar and PDF were created exclusively.
- Tip `/workspace/residual-c-field-WRITTEN_PREMISES_DRAFT.md` sha256 = 7cddb188ca340ad977516327065e8df88c3b16ed01bcd8301c5bf0f8bb5dfc91, verified at the start and re-verified after writing.
- No CERT. No Gemini. Nothing sent outside.